

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

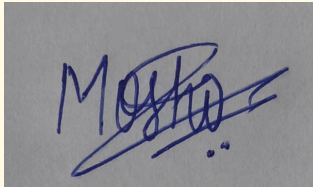
Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA0305	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
Student Name:	Moshika M	Reg. No.:	192424064
Section / Batch:	B.Tech (AI-DS)	Date:	13.08.2026

Code of Conduct

I Moshika M - 192424064 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the Student:

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

1. Semantic Representation in Customer Support Chatbots.

A telecom company collected customer chatbot interactions.
The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

1. Analyze the semantic representations and identify the action-object relationships.
2. Determine which query contains semantic interpretation errors.
3. Evaluate how meaning representation affects chatbot decision accuracy.
4. Recommend improvements to enhance semantic understanding in telecom customer support systems.

ANSWER:

1. Analysis of Action-Object Relationships

Semantic representation identifies the main action performed by the customer and the object or service affected by that action. The given representations follow the general form **ACTION (Object, Customer)**.

- **ACTIVATE (Roaming, Customer)**: The customer wants to activate international roaming. The action is **ACTIVATE**, and the object is **Roaming**.
- **DEACTIVATE (CallerTune, Customer)**: The customer wants to stop the caller tune service. The action is **DEACTIVATE**, and the object is **CallerTune**.
- **QUERY (DataBalance, Customer)**: The customer is requesting information about their available data balance. The action is **QUERY**, and the object is **DataBalance**.
- **ACTIVATE (5GService, Customer)**: The customer wants to enable the 5G service. The action is **ACTIVATE**, and the object is **5GService**.

Thus, semantic representations convert natural-language queries into structured meanings that can be processed by the chatbot.

2. Identification of Semantic Interpretation Errors

From the chatbot logs:

- **Q1:** Actual = Activate Roaming, Predicted = Activate Roaming → **Correct**
- **Q2:** Actual = Deactivate Caller Tune, Predicted = Activate Caller Tune → **Incorrect**
- **Q3:** Actual = Check Data Balance, Predicted = Query Data Balance → **Correct**
- **Q4:** Actual = Enable 5G Service, Predicted = Activate 5G Service → **Correct**

The major semantic error occurs in **Q2**. The system identified the correct object, Caller Tune, but assigned the wrong action. The user's intention was to **DEACTIVATE** the service, whereas the chatbot interpreted it as **ACTIVATE**. This could result in the system enabling a service that the customer actually wanted to disable.

3. Effect of Meaning Representation on Chatbot Accuracy

Meaning representation directly influences chatbot decision-making. If the semantic analyzer correctly identifies both the action and object, the chatbot can select the appropriate backend operation.

For example:

"Enable 5G service" → ACTIVATE(5GService, Customer)

The chatbot can therefore trigger the activation procedure.

However, if the representation is incorrect:

"Deactivate caller tune" → ACTIVATE(CallerTune, Customer)

the chatbot may perform the opposite operation. Therefore, errors in semantic representation can lead to incorrect service execution, customer dissatisfaction, and potentially unwanted charges.

The four queries contain three correct interpretations, giving an intent-level accuracy of **75%**. This demonstrates that even a small number of semantic errors can affect the reliability of an automated customer-support system.

4. Recommended Improvements

The telecom chatbot can be improved using the following methods:

1. **Context-aware intent classification:** Analyze surrounding words and conversation history to distinguish between activation and deactivation requests.
2. **Action-object validation:** Verify that the identified action is compatible with the detected service or object before executing an operation.
3. **Synonym and paraphrase handling:** Recognize expressions such as *enable*, *activate*, *turn on* and *disable*, *deactivate*, *turn off* as related action groups.
4. **Confidence-based confirmation:** If the system has low confidence, it should ask the customer for confirmation before performing the operation.
5. **Continuous model training:** Use historical chatbot conversations and corrected user interactions to improve semantic classification.

Overall, accurate meaning representation is essential for reliable telecom chatbots because it connects the customer's natural-language request with the correct system operation.

2. First-Order Predicate Calculus for Smart Manufacturing.

An Industry 4.0 manufacturing plant monitors machines using semantic rules.

Production Data:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- $\text{Active}(x) \rightarrow \text{Producing}(x)$
- $\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$
- $\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

Tasks:

1. Represent the production data using First-Order Predicate Calculus.
2. Analyze which products are currently available.
3. Infer whether Gear production is affected by maintenance activities.
4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.

ANSWER:

1. Representation of Production Data Using First-Order Predicate Calculus

The production status of each machine can be represented using predicates.

Given:

- M1 is Active: $\text{Active}(M1)$
- M2 is Active: $\text{Active}(M2)$
- M3 is under Maintenance: $\text{Maintenance}(M3)$
- M4 is Active: $\text{Active}(M4)$

The general production rules can be represented as:

- $\text{Active}(x) \rightarrow \text{Producing}(x)$
- $\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$
- $\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

In first-order logic notation, the rules are:

$\forall x [\text{Active}(x) \rightarrow \text{Producing}(x)]$

$\forall x \forall y [(\text{Produces}(x,y) \wedge \text{Active}(x)) \rightarrow \text{Available}(y)]$

$\forall x [\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)]$

From the given facts, we can infer:

- $\text{Producing}(M1)$
- $\text{Producing}(M2)$

- $\neg \text{Producing}(\text{M3})$
- $\text{Producing}(\text{M4})$

2. Products Currently Available

The second rule states that if an active machine produces a product, that product is available:

$\text{Produces}(\text{x}, \text{y}) \wedge \text{Active}(\text{x}) \rightarrow \text{Available}(\text{y})$

Therefore, products produced by **M1**, **M2**, and **M4** are considered available because these machines are active.

However, the problem does not explicitly specify which products each machine produces. Therefore, specific product names cannot be logically determined from the provided data alone.

If, for example, **M1 produces Gear**, then:

$\text{Produces}(\text{M1}, \text{Gear}) \wedge \text{Active}(\text{M1})$

allows us to infer:

$\text{Available}(\text{Gear})$

3. Effect of Maintenance on Gear Production

M3 is under maintenance: $\text{Maintenance}(\text{M3})$

Using the rule:

$\text{Maintenance}(\text{x}) \rightarrow \neg \text{Producing}(\text{x})$

we can infer: $\neg \text{Producing}(\text{M3})$

Therefore, if **M3 is responsible for producing Gear**, Gear production is affected and temporarily stopped.

However, if Gear is produced by M1, M2, or M4, production can continue because those machines are active. Hence, the effect on Gear depends on which machine is responsible for producing it.

4. Effectiveness of Predicate Logic in Industrial Decision Support

Predicate logic is useful in smart manufacturing because it represents machines, products, conditions, and relationships in a formal manner. It allows the system to automatically derive new information from existing production facts and rules.

For example, when a machine changes from **Active** to **Maintenance**, the system can infer that it is no longer producing. Similarly, if an active machine produces a particular product, the system can determine that the product is available.

Predicate logic provides transparency because each conclusion can be traced back to a specific rule and fact. It can therefore support production monitoring, machine-status analysis, availability checking, and automated decision-making.

However, real manufacturing environments contain uncertainty, sensor errors, and changing conditions. Therefore, predicate logic can be combined with probabilistic or machine-learning techniques for more complex industrial applications.

3. Word Sense Disambiguation in E-Commerce Search Engines.

An e-commerce company observed the following search logs.

User Query	Possible Sense 1	Possible Sense 2
"Apple accessories"	Fruit	Technology Brand
"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language
"Python course"	Snake	Programming Language

Search Context Data:

Query	Clicked Result
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.
4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

ANSWER:

1. Determining the Correct Word Sense

Word Sense Disambiguation (WSD) determines the intended meaning of an ambiguous word by analyzing its context.

The correct senses are:

- **Apple accessories** → **Technology Brand**: The clicked result is *iPhone Charger*, indicating that "Apple" refers to the technology company rather than the fruit.
- **Mouse wireless** → **Computer Device**: The clicked result is *Bluetooth Mouse*, showing that "mouse" refers to a computer peripheral rather than an animal.
- **Java tutorial** → **Programming Language**: The result *Coding Lessons* indicates that Java refers to the programming language.
- **Python course** → **Programming Language**: The result *Software Development Training* indicates that Python refers to the programming language.

Therefore, all four queries are correctly disambiguated using their search context and clicked results.

2. Semantic Cues Used for Disambiguation

Several cues help identify the intended meaning:

- **Associated words:** Words such as *accessories*, *wireless*, *tutorial*, and *course* provide contextual information.
- **Domain-specific vocabulary:** Terms such as *iPhone*, *Bluetooth*, *coding*, and *software development* strongly indicate technology-related meanings.
- **User interaction:** The clicked search result provides evidence about what the user intended.
- **Query structure:** The combination of ambiguous words with other terms helps determine the intended sense.

For example, the word **Java** alone is ambiguous, but "**Java tutorial**" strongly suggests programming because *tutorial* commonly occurs with programming-related searches.

3. Impact of Incorrect Sense Selection

Incorrect WSD can significantly reduce search-engine performance. If the system interprets "**Apple accessories**" as fruit-related, it might display fruit baskets or kitchen products instead of iPhone chargers and other Apple accessories.

Similarly, interpreting "**Python course**" as a snake-related query could produce irrelevant wildlife information rather than programming courses.

Incorrect sense selection can lead to:

- Irrelevant search results.
- Lower click-through rates.
- Poor recommendation accuracy.
- Increased user frustration.
- Reduced customer engagement.
- Lower trust in the search engine.

Therefore, accurate WSD is important for improving both search relevance and the overall user experience.

4. Industrial-Scale WSD Strategy

An e-commerce company can implement a hybrid WSD system combining multiple sources of information.

1. **Contextual language models:** Use transformer-based models to understand the surrounding words and meaning of ambiguous terms.
2. **Domain knowledge:** Maintain product catalogs, category information, and knowledge graphs to distinguish technology, food, animals, and other domains.
3. **User behavior:** Use clicks, purchases, search history, and browsing behavior as additional evidence for identifying user intent.
4. **Query expansion:** Expand short queries using related terms and product categories to improve semantic understanding.
5. **Continuous learning:** Retrain and update the model using new queries, clicks, purchases, and user feedback.

A large-scale WSD system can therefore combine linguistic context, product information, and behavioral signals to select the most probable meaning. This improves search relevance and recommendation accuracy.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	Role
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

Tasks:

1. Analyze how syntactic structures contribute to semantic role assignment.
2. Determine whether the assigned semantic roles are appropriate.
3. Evaluate potential errors that could arise from incorrect parsing.
4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

ANSWER:

1. Relationship Between Syntax and Semantic Roles

Syntactic analysis determines how words are grammatically related within a sentence. These relationships help the NLP system assign semantic roles such as **Agent**, **Patient**, **Instrument**, **Recipient**, and **Symptom**.

For example:

"Doctor prescribed medicine to patient."

The syntactic structure identifies:

- Doctor → Subject
- prescribed → Verb
- medicine → Object
- patient → Indirect object/recipient

From this structure, the system can assign:

- **Doctor** → **Agent**
- **Medicine** → **Theme**
- **Patient** → **Recipient**

Similarly:

"Patient reported severe headache."

The patient is the **Agent/Experiencer**, while headache represents the **Symptom** being reported.

Thus, syntactic relationships provide an important foundation for understanding the meaning of medical statements.

2. Appropriateness of the Assigned Semantic Roles

Some of the given semantic roles are appropriate, while others need correction.

- **Doctor** → **Agent**: Appropriate because the doctor performs the action of prescribing.
- **Medicine** → **Instrument**: Generally inappropriate. Medicine is more accurately represented as the **Theme/Object** of the prescribing action.
- **Patient** → **Recipient**: Appropriate in the sentence *"Doctor prescribed medicine to patient."*
- **Headache** → **Symptom**: Appropriate because it represents the patient's reported medical symptom.

For **"Medicine reduced blood pressure,"** medicine is the entity causing the reduction and can be considered the **Cause/Agent-like role**, while blood pressure is the affected entity or **Theme**.

Therefore, semantic-role assignment should consider the meaning of the verb and the relationship between entities, rather than relying only on grammatical position.

3. Errors Caused by Incorrect Parsing

Incorrect syntactic parsing can lead to incorrect semantic interpretation. In healthcare applications, this can be particularly serious.

For example, if the system incorrectly identifies the patient as the subject of a sentence, it may assign the wrong role and misunderstand who performed an action.

Potential errors include:

- Incorrect identification of the doctor, patient, or medication.
- Confusing the person receiving treatment with the person providing treatment.
- Misinterpreting symptoms as diseases or treatments.
- Incorrectly associating medications with patients.
- Missing relationships between medical events.

Such errors could affect clinical information extraction, medical summarization, decision-support systems, and electronic health-record analysis.

4. Methods to Improve Syntax-Driven Semantic Analysis

Healthcare NLP systems can be improved through the following approaches:

1. **Medical-domain parsers:** Use syntactic models trained specifically on clinical and biomedical text.
2. **Dependency parsing:** Analyze relationships between words rather than relying only on simple Subject-Verb-Object structures.
3. **Named Entity Recognition:** Identify doctors, patients, medicines, diseases, symptoms, and procedures accurately.
4. **Semantic Role Labeling:** Use context-aware models to assign roles based on the meaning of each medical verb.
5. **Medical knowledge bases:** Integrate clinical ontologies and medical terminology databases to validate extracted relationships.

Overall, syntax-driven semantic analysis provides an important foundation for healthcare NLP, but syntactic information should be combined with semantic context and medical knowledge to achieve accurate interpretation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____